Petroleum College International Killsheet

W & W or Drillers Method

Name:	Well Information	Well Name:

Pump Info	1		2	3	Well Depth:		4
Output	bbls/ stks	stks/min =	SCRP psi	Current mud		MD	TVD
Surface line volume barrels =				6	CSG Depth:		5
DP capacity				Test mud	Shoe	MD	TVD
	bbls/ft	X length MD =	Barrels in DP	7			
HWDP capacity				Surface test pressure FIT/LOT			
	bbls/ft	X length MD =	Barrels /HWDP				
DC capacity							
	bbls/ft	X length MD =	Barrels in DC				

Various names used: Current mud = Original or Present mud.

Slow circulating rate pressure (SCRP) = Kill rate pressure or slow pump pressure. Slow pump rate = Kill rate speed.

Total Barrels in Drill String		1	=		
		÷ Pump Output bbls/stk	=	**Total Strokes Surface to Bit**	

Annular capacity between:			
DP & Csg			
Volume	bbls/ft	X length DP/Csg =	Barrels
DP & OH			8
Volume	bbls/ft	X length DP/OH =	Barrels
HWDP & OH			9
Volume	bbls/ft	X length DP/OH =	Barrels
DC & OH			10
Volume	bbls/ft	X length DC/OH =	Barrels
Choke line vol.			
(Subsea only)	bbls/ft	X length MD =	Barrels

Strokes bit to casing shoe

8	9	10	11
bbls between DP & OH	+ bbls between HWDP & OH	+ bbls between DC & OH	= Volume Bit to Csg shoe

11	1	
Volume Bit to Csg shoe	÷ Pump output bbls/stk	= Strokes Bit to Csg shoe

Formulas:	**Total Barrels in Annulus**		1	=	
			÷ Pump output bbls/stk	=	**Total Stroke Bit to Surface**

Pipe capacity = $ID^2 \div 1029.4$ = bbls/ft

Annular capacity = $(ID^2 - OD^2) \div 1029.4$ = bbls/ft

Total Well Volume Drill String & Annulus		**Total Strokes Surface to Surface**		**String & Annulus**

Well Kill Calculations & Pressure Considerations

SIDPP	12	SICP		KICK SIZE		(Subsea only) Slowly reduce CLFP off casing bringing pump on line to SCR.
Kill Weight Mud:	12	0.052	4	3	13	
	(SIDPP	÷ 0.052	÷ Well depth TVD)	+ Current mud =	KWM	
Initial Circulating Pressure:		12	2	=	14	
		SIDPP	+ SCRP	=	ICP (Initial Circulating Pressure)	
Final Circulating Pressure:		2	13	3	15	
		SCRP	X KWM	÷ Current mud	= FCP (Final Circulating Pressure)	
Max. Allowable Mud Wt:	7	0.052	5	6	16	
MAMW: LOT/FIT	(Surface test pressure ÷	0.052	÷ Csg shoe TVD) +	Test mud	= MAMW	
Max. Allowable Annular	16	3	0.052	5		
Surface Pressure MAASP:	(MAMW	- Current mud)	X 0.052	X CSG shoe TVD =	MAASP	
New Max. Allowable Annular	16	13	0.052	5		
Surface Pressure after Kill:	(MAMW	- KWM)	X 0.052	X CSG shoe TVD =	New MAASP Kill Fluid	

ICP to FCP Pressure Reduction Schedule Surface to Bit

Stks surf to bit	0										Bit
DP calculated	14 ICP										15 FCP
mins or bbls or gals											

Pump schedule: Strokes to bit ÷ 10 = Strokes increase per step (ICP - FCP) ÷ 10 = Pressure reduction per step